

MeshSmooth

The MeshSmooth modifier is the most common way of subdividing a mesh. The modifier has several algorithms that it can use to smooth out the mesh. Classic smoothens the mesh to three- and four-sided faces, while Quad Output outputs only four-sided faces. The most robust and commonly used method is NURMS (non-uniform rational MeshSmooth), which allows for much more sophisticated control, such as weighting each control vertex. MeshSmooth also has the ability to apply smoothing in levels, so that part of an object can be smoothed differently than the rest. Iterations control how much the mesh is subdivided. You can select different values for the iterations shown in the viewports and for the iterations that are actually rendered. Because the number of polygons in the object quadruples with each iteration, be careful about using high numbers for iterations.

MeshSmooth allows you to see the underlying polygonal object as a “cage” superimposed on the smoothed object. The vertices and edges of this cage can be selected as sub-objects for the purposes of weighting, which can create creases or sharp edges.

TurboSmooth

TurboSmooth is a simpler way to subdivide surfaces and will do much of the work of MeshSmooth without the overhead. It is much faster and more memory-efficient than MeshSmooth. It simply subdivides the mesh and allows you to set the iterations seen in the viewport and at rendering time. TurboSmooth also has an option for Explicit Normals, unavailable in MeshSmooth.

Smoothing editable polys

An editable poly object also has the ability to subdivide a surface. This option is available only in the editable poly object, but not in the Edit Poly modifier, so in order for this to be used, the stack must be collapsed to Editable Poly. The smoothing algorithm is based on MeshSmooth, and the controls are a subset of the modifier.

Because the smoothing, using Editable Poly happens at the bottom of the stack and works well only for certain types of objects such as those that do not need additional modifiers in the stack. For applications such as character animation, the character is typically smoothed after skin and other deformation modifiers are added to the stack, so smoothing in the base object would not be applicable. 